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Rigid cutterbar over travel

Abstract

A cutter bar assembly for an agricultural vehicle header includes a cutter bar supported by movable arms extending from a header frame and capable of operation in flex and rigid modes. The support arms are joined to the header frame such that they may be displaced from a raised position in the rigid mode when subjected to a lifting force due to striking an obstacle without damage to the header structure. Also, an agricultural vehicle header and an agricultural combine harvester includes the cutter bar assembly.

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Background/Summary

FIELD OF THE INVENTION

(1) The present invention is related to combine harvesters, in particular to a header for a combine harvester, more particularly to a mechanism for protecting the cutterbar and infeed of a combine header against damage from striking obstacles in a rigid operating mode.

BACKGROUND OF THE INVENTION

(2) Agricultural combine harvesters are vehicles for harvesting and processing crops such as wheat or corn. A combine harvester typically includes a laterally-extending header that is movably attached to the front of the combine chassis. Crops are cut from the field by a cutter bar extending across the front of the header. An auger or a belt (draper) system brings the cut material to a central area of the header, and from there an infeed mechanism transports the cut crops to the interior of the vehicle to a threshing mechanism and further to a cleaning section, where grains are separated from crop residue.

(3) The header typically is located at the front of the vehicle and extends laterally relative to the vehicle's direction of travel. The header can be a single rigid body, or it can be a multi-segment or articulated header, comprising multiple sections that are movable relative to each other and to the vehicle chassis. Hydraulic or pneumatic cylinders, screws, springs, or other actuators, linkages, and the like may be provided to selectively control the positions of the header and its constituent parts. During operation, the header and/or cutter bar might be raised or lowered to account for variations in the ground level, properties of the particular crop being harvested, or various other operating conditions or requirements.

(4) Thus, a typical header might be pivotally mounted to the vehicle chassis by way of a feeder housing that can be moved up and down to raise and lower the entire header. The header also may include a forward tilt adjustment mechanism that rotates the entire header forward and backward to change the angle of the header relative to the ground. Still further, the header might have a lateral tilt adjustment mechanism to rotate the header about the fore-aft axis to account for different ground levels in the lateral direction. A header having height and tilt adjust mechanisms may be operated by actively raising and lowering the header to account for undulations in the ground. Ground distance sensing equipment located below the header signals the header height position controller to raise and lower the entire mass of the header to avoid damaging contact with irregular ground or obstacles.

(5) The cutter bar is disposed forward of the header frame on movable supports that allow the cutter bar to move relative to the header frame. Various cutter bar support systems are known in the art. For example, the cutter bar may be mounted on support arms that are mounted on the header frame by a pivot connection. Still further, the cutter bar may comprise a flexible cutter bar that is supported by multiple independently-movable supports along its lateral extent, allowing the cutter bar to flex to conform more precisely to lateral undulations in the terrain. In a typical arrangement, a support arm is pivotably attached at its aft or proximal end to the header main frame, and a stop link is provided between the support arm and the header frame to limit upward and downward

movement of the support arm relative to the frame.
(6) For certain harvesting operations requiring cutting crops close to the ground, the support arms are freely movable relative to the header frame, and the cutter bar flexes up and down on support arms tracking the local shape of the ground below each support arm (flex mode). For other operations, it is desirable or necessary to secure or lock the movable supports to fix the cutter bar in a stationary position with respect to the header main frame (rigid mode). For example, it is common to lock the cutter bar support arms in a raised position during transport of the harvester over fields and other terrain, to prevent the cutter bar from bouncing or from striking the ground or other terrestrial obstacles encountered by the moving harvester. In still other operations, it may be desirable to lock the cutter bar at an intermediate or even a completely lowered position. To facilitate locking of the cutter bar, the support arms may be provided with stop links that limit their upward or downward movement.
(7) When the cutter bar in such a system is locked in a raised position in the rigid mode, the support arm cannot move upward any further relative to the header. In the center section of the header, where the infeed frame acts as the support arm for the cutter bar, there is an area where the ground or obstacles may not be sensed or may be sensed only poorly by the header position control system. If an obstacle is encountered near the infeed and not detected as the header passes over, and the control system does not react to lift the header, contact between the locked cutter bar or header infeed section and the obstacle may occur to the point where damage is inflicted on the header structure. This damage can be minimal (bending mounting pins) or major (bending, e.g., the entire infeed frame). While the state of the art provides various alternatives to avoid header damage due to unintended ground contact, there remains a need to advance the state of the art.
(8) This description of the background is provided to assist with an understanding of the following explanations of exemplary embodiments, and is not an admission that any or all of this background information is necessarily prior art.
SUMMARY OF THE INVENTION
(9) In one exemplary aspect, there is provided a cutter bar assembly for an agricultural vehicle header having a header frame, at least one support arm extending from the header frame in a forward direction, the support arm joined to the header frame at an end proximal to the header frame by a pivot, a cutter bar attached to a respective distal end of each at least one support arm, and a stop link. The stop link has first and second ends, wherein the first end joins one of the header frame or the support arm by a pivot and the second end joins the other of the header frame or the support arm by a sliding joint, and wherein the stop link joins the header frame at a point distal from the pivot joining the support arm to the header frame. A hydraulic actuator is joined at one end to the header frame by a pivot and at an opposite end to the support arm by a sliding joint, wherein the actuator can lift and hold the support arm in a raised position in a rigid mode. The sliding joint joining the hydraulic actuator to the support arm and the stop link enable the support arm to move upward from the raised position in the rigid mode to the over-traveled position when the support arm is subjected to a lifting force at its distal end.
(10) In some exemplary aspects, the at least one support arm is or forms a part of an infeed frame supporting the infeed section of a header.
(11) In some exemplary aspects, the sliding joint of the stop link is a pin slot joint formed by a slot in the stop link and a pin fixed respectively either to the header frame or the support arm. The stop link may be joined by a pivot to the support arm and by a sliding joint to the header frame. The stop link may be joined to the header frame at a point below the pivot joining the support arm to the header frame.
(12) In some exemplary aspects, the sliding joint joining the actuator to the support arm is a pin slot joint formed by a pin fixed to the actuator and a slot fixed to the support arm.
(13) In some exemplary aspects, the cutter bar assembly further includes an actuator link joined to the header frame and support arm in parallel with the hydraulic actuator to limit travel of the

hydraulic actuator. The actuator link may be joined to the header frame and support arm by at least one sliding joint. The sliding joints may be formed by a slot in the actuator link and a pin fixed respectively to either or both of the header frame and the support arm.
(14) In some exemplary aspects, the stop link may further incorporate a spring or spring mechanism. The stop link may also incorporate a strain gauge sensor or a load cell sensor for detecting when the support arm is subjected to a lifting force at its distal end.
(15) In some exemplary aspects, the support arm incorporates a position sensor for detecting when the support arm is moved upward from the raised position in the rigid mode by a lifting force at its distal end.
(16) In some exemplary aspects, there is further provided an agricultural vehicle header incorporating the cutter bar assembly. The header may be provided with a header position control system, and the cutter bar assembly may be provided with a position sensor for detecting when the support arm is moved upward from the raised position in the rigid mode by a lifting force at its distal end and signalling the header position control system to raise the header. In some exemplary embodiments, there is provided an agricultural combine harvester incorporating the agricultural vehicle header.
(17) In some exemplary aspects, there is provided a cutter bar assembly for an agricultural vehicle header having a header frame, an infeed frame extending from the header frame in a forward direction, the infeed frame joined to the header frame at an end proximal to the header frame by a pivot, and a cutter bar attached to a respective distal end of the infeed frame A stop link is joined by a pivot to the infeed frame and by a pin slot joint to the header frame, wherein the pin slot joint comprises a slot in the stop link and a pin fixed to the header frame at a point below the pivot joining the infeed frame to the header frame. A hydraulic actuator is joined at one end to the header frame by a pivot and at an opposite end to the support arm by a sliding joint formed by a pin slot joint formed by a pin fixed to the actuator and a slot fixed to the infeed frame, wherein the actuator can lift and hold the infeed frame in a raised position in a rigid mode. The sliding joint joining the hydraulic actuator to the infeed frame and the stop link enable the infeed frame to move upward from the raised position in the rigid mode when the infeed frame is subjected to a lifting force at its distal end. The cutter bar assembly further has an actuator link joined by sliding joints to the header frame and infeed frame in parallel with the hydraulic actuator to limit travel of the hydraulic actuator, wherein the sliding joints joining the actuator link to the header frame and infeed frame are formed by a slot in the actuator link and a pin fixed respectively to each of the header frame and the infeed frame. The cutter bar assembly further has a position sensor for detecting when the infeed frame is moved upward from the raised position in the rigid mode by a lifting force at its distal end. In a further exemplary aspect, there is provided an agricultural vehicle header, and the infeed frame position sensor signals the header position control system to raise the header when the position sensor detects the infeed frame is moved upward from the raised position in the rigid mode by a lifting force at its distal end.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The foregoing summary, as well as the following detailed description of several aspects of the subject application, will be better understood when read in conjunction with the appended drawings. For the purpose of illustrating the subject application there are shown in the drawings several aspects, but it should be understood that the subject application is not limited to the precise arrangements and instrumentalities shown. Embodiments of inventions will now be described, strictly by way of example, with reference to the accompanying drawings, in which:
(2) FIG. **1** is a side view of an agricultural combine harvester.

(3) FIG. **2** is an isometric view of a header.
(4) FIG. **3** is an isometric view of a portion of a header frame.
(5) FIG. **4** is an isometric view of a portion of an exemplary embodiment of the cutter bar assembly.
(6) FIG. **5** is an isometric detail of a portion of an exemplary embodiment of the cutter bar assembly, showing the connection of the hydraulic actuator and actuator link to the infeed frame.
(7) FIG. **6** is a side view of an exemplary embodiment of the cutter bar assembly, showing the infeed frame in the rigid position, including a breakout detail view of the connection of the hydraulic actuator to the infeed frame.
(8) FIG. **7** is a side view of an exemplary embodiment of the cutter bar assembly, showing the infeed frame in the over-traveled position, including a breakout detail view of the connection of the hydraulic actuator to the infeed frame.
(9) FIG. **8** is a side view of an exemplary embodiment of the cutter bar assembly, showing the infeed frame in the rigid and over-traveled positions, including a breakout detail view of the connection of the hydraulic actuator to the infeed frame.
(10) FIG. **9** is an isometric detail of a portion of an exemplary embodiment of the cutter bar assembly, showing the stop link in the rigid operating mode while the infeed frame is in the raised position.
(11) FIG. **10** is an isometric detail of a portion of an exemplary embodiment of the cutter bar assembly, showing the stop link in the flex operating mode while the infeed frame is in the fully lowered position.
(12) FIG. **11** is an isometric detail of a portion of an exemplary embodiment of the cutter bar assembly, showing the stop link in the over-traveled position.
(13) FIG. **12** is an isometric detail of an exemplary embodiment of the cutter bar assembly, showing the infeed frame position sensor.
(14) FIG. **13** is a schematic view of a control system.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS OF THE INVENTION

(15) The foregoing summary, as well as the following detailed description of several aspects of the subject application, will be better understood when read in conjunction with the appended drawings. For the purpose of illustrating the subject application there are shown in the drawings several aspects, but it should be understood that the subject application is not limited to the precise arrangements and instrumentalities shown. Exemplary embodiments of the present invention provide cutter bar and draper support assemblies for headers for agricultural equipment, such as combines, swathers, windrowers, and the like. It will be appreciated that other embodiments may be used in other types of machines having a similar arrangement of parts, upon incorporation of the appropriate features of the inventions herein. The detailed description is not limiting the scope of the invention, which is defined only by the appended claims.
(16) Preferred embodiments will now be described with reference to the drawings. The terms ‘front’ and ‘back’ or ‘rear’ are referenced to the front and back side of the agricultural combine harvester. The ‘forward direction’ of the agricultural combine harvester refers not to a single geometrical axis but to the general direction from the rear of the vehicle to the front.
(17) FIG. **1** illustrates an example of an agricultural combine **100**, with which embodiments of the invention may be used. The combine **100** includes a chassis **102** that is configured for driving on a surface (e.g., the ground or a road), such as by being supported by pneumatic wheels **104**, tracked wheel assemblies, or the like. The combine **100** includes a threshing and separating system **106** mounted on or within the chassis **102**. The threshing and separating system **106** may include mechanisms such as one or more threshers (e.g., an axial flow thresher), sieves, blowers, and the like, as well as an associated grain hopper and unloader. Threshing and separating systems **106** and their associated components are well-known in the art, and need not be described in detail herein. The combine **100** also may include other features, such as a spreader **108**, operator cab **110**, and the

like.

(18) Referring also to FIGS. **2** and **3**, the combine **100** also includes a header **112**, which is configured to cut and harvest crop material from the ground as the combine **100** drives in the forward direction F. For example, the header **112** may include one or more cutter bars **114** located at or near the leading edge of the header **112** to cut crops at or near the ground level, and one or more reels **116** configured to pull the crop material backwards towards the header **112**. The header **112** also may include crop conveyors **118** that are configured to move the crop material at the lateral ends of the header **112** towards the center of the header **112**. The crop conveyors **118** may be in the form of belts, auger screws, or the like. At the center, the header **112** may include a infeed section **140** having a feeder conveyor **120** that conveys the crop material backwards towards a crop outlet **122**. The header **112** also may include gauge wheels **124** or skids to control the height of the header **112** over the ground.

(19) The header **112** is built on a frame **126**, which is attached to the chassis **102** by a feeder housing **128**. The feeder housing **128** is configured to convey crop material backwards from the header **112** to the threshing and separating system **106**. The feeder housing **128** may be movable by one or more feeder housing actuators **130** to raise and lower the header **112** in a vertical direction V relative to the ground.

(20) The illustrated exemplary header **112** is a unitary header having a single frame that extends continuously between the ends of the header **112** in the lateral direction L. In other embodiments, the header **112** may comprise a multi-segment or articulated header having a center section and one of more wing sections movably attached to the lateral end of the center section by pivots or linkages. In either case, the header **112** has a central region defined by the lateral extent of the crop outlet, and lateral regions extending in the lateral direction L from the central region.

(21) The header **112** also includes a number of support arms **132** that extend forward from the frame **126** to hold parts such as the cutter bar **114**, conveyors **118**, or the like. The support arms **132** may be rigidly attached to the header **112**, or attached by movable mounts, such as pivots or linkages. In the case of movable support arms **132**, a suspension may be used to control the motion of the support arms **132**. For example, each support **132** may have its own spring and/or damper system, which is intended to allow the support arms **132** to move up and down individually or in groups to follow local undulations along the lateral direction L. At the center of the header, the infeed frame **142** acts as the support arm for the cutter bar **114**, and is similarly attached to the frame **126** and controlled as are the support arms **132**. Skids, gauge wheels or other ground supports may be located below the support arms **132** to generate a lifting force via contact with the ground. The positions of the ground supports and the spring and damping properties of the movable connections may be adjustable to tailor the header **112** for use in particular operating conditions. In addition, the positions of the support arms **132** and infeed frame **142**, such as their angular orientation (downward tilt) relative to the frame **126**, may be adjustable.

(22) In use, the header **112** cuts crop materials with the cutter bar **114**, receives the crop materials on the conveyors **118**, **120**, and conveys the crop materials back through the crop outlet **122** to the threshing and separating system **106**. The movement of the crop materials in the lateral regions of the header **112** is controlled, in part, by one or more backsheets **136**. The backsheets **136** are located in the lateral regions of the header, at rear edges of the lateral conveyors **118**, and prevent crop materials from falling behind the lateral conveyors **118**. The backsheets **136** may slope backwards, as shown in FIG. **2**, or they may be vertical or have other profiles.

(23) In the example of FIG. **3**, it can be seen that the support arms **132** and the infeed frame **142** extend forward from a main truss **138** of the header frame **126**. The main truss **138** is located at a lower end of the frame **126**, and extends in the lateral direction L. Thus, the main truss **138** acts as the structural backbone of the frame in holding the support arms **132** and infeed frame **142**. Each support arm **132** and the infeed frame **142** are pivotally connected to the header frame **126** at a respective pivot **134**.

(24) For certain harvesting operations requiring cutting crops close to the ground, the support arms **132** and infeed frame **142** are freely movable relative to the header frame **126** in a state known as the flex mode, and the cutter bar **114** flexes up and down on the support arms **132** and infeed frame **142**, tracking the local shape of the ground below each support arm **132** or infeed frame **142**. For other operations, it is desirable or necessary to secure or lock the movable supports **132** and **142** to fix the cutter bar **114** in a stationary position with respect to the header frame **126** in a state known as the rigid mode. For example, the cutter bar support arms **132** and infeed frame **142** might be locked in a fully raised position during transport of the agricultural harvester **100** over fields and other terrain, to prevent the cutter bar **114** from bouncing or from striking the ground or other terrestrial obstacles that may be encountered in transit.
(25) Referring now to FIG. **4**, a first example of a cutter bar assembly for an agricultural vehicle header is shown. The cutter bar assembly includes a header frame **126** and at least one support arm **132** extending from the header frame **126** in a forward direction. The support arm **132** is joined to the header frame **126** at an end proximal to the header frame **126** at pivot **134**. In the embodiment shown in FIG. **4**, the support arm **126** supports the cutter bar **114**, but also functions as the infeed frame **142**, supporting the feeder conveyor **120**, only the left side of which appears in the detail of FIG. **4**; a right side of the infeed frame **142** similarly extends from the header frame **126** and joins the header frame at a respective pivot **134**. The cutter bar **114** (not shown in FIG. **4**) is attached to a respective distal end of the support arm **132** or infeed frame **142**. For purposes of the cutter bar assembly, the support arm **132** and infeed frame **142** are interchangeable.
(26) At a first end **404**, a stop link **400** joins the header frame **126** to the support arm **132**/infeed frame **142** at a point **402** distal from the pivot **134** joining the support arm **132**/infeed frame **142** to the header frame **126**. A second end **406** of the stop link **400** joins the support arm **132**/infeed frame **142** at point **408**. The first end **404** of the stop link **400** may be joined to the header frame **126** at point **402** by a sliding joint, in which case the second end **406** of the stop link **400** is joined to the support arm **132**/infeed frame **142** by a pivot. Alternately, the second end **406** of the stop link **400** may be joined to the support arm **132**/infeed frame **142** at point **408** by a sliding joint, in which case the first end **404** of the stop link **400** would be joined to the header frame **126** at point **402** by a pivot. In either configuration, the sliding joint of the stop link **400** defines a range of motion for the support arm **132**/infeed frame **142** as is rotates at pivot **134**.
(27) A hydraulic actuator **410** is joined at first end **412** to the header frame **126** by a pivot **414** attached to the header frame **126** by support arm **416** and at a second end **418** to the support arm **132**/infeed frame **142** by a sliding joint **420**. The hydraulic the actuator can lift and hold the support arm **132**/infeed frame **142** in a raised position in a rigid mode, wherein the sliding joint joining the hydraulic actuator **410** to the support arm **132**/infeed frame **142** and the sliding joint of the stop link **400** enable the support arm **132**/infeed frame **142** to move upward from the raised position in the rigid mode when the support arm **132**/infeed frame **142** is subjected to a lifting force distal from the infeed frame **126**. An actuator link **422** is joined to the header frame **126** by support arm **416** and to the support arm **132**/infeed frame **142** in parallel with the hydraulic actuator **410** for limiting the extension or compression travel of the hydraulic actuator **410** to prevent damage to the actuator from over-extension or over-compression.
(28) FIG. **5** shows the arrangement of the hydraulic actuator **410** and the actuator link **422** in greater detail. The hydraulic actuator **410** is joined to the support arm **132**/infeed frame **142** by a sliding joint **420**. The sliding joint **420** is a pin slot joint formed by a slot **500** fixed to the support arm **132**/infeed frame **142** and a pin **502** fixed to the second end **418** of the hydraulic actuator **410**. The hydraulic actuator **410** is joined to the header frame **126** by a pivot **414** fixed to header frame **126** by support arm **416**. In alternative embodiments (not shown), sliding joint **420** may comprise an open slot, rather than the closed slot shown, or a linkage, such as a parallel linkage.
(29) The actuator link **422** is joined to the header frame **126** by a sliding joint formed by a slot **504** in the actuator link **422** and a pin **506** fixed to the header frame **126** by the support arm **416**. The

actuator link **422** is joined to the support arm **132**/infeed frame **142** by a pivot **508**. In an alternative arrangement (not shown), the actuator link **422** can be attached to the support arm **132**/infeed frame **142** by a sliding joint and to the header frame **126** by a pivot; in a further alternative arrangement (also not shown), the actuator link **422** can be attached to both the support arm **132**/infeed frame **142** and the header frame **126** by a sliding joint. In all arrangements the actuator link **422** limits the travel of the actuator **410** to prevent damage by over-compression or over-extension.

(30) FIGS. **6**, **7**, and **8** show the cutter bar assembly in various modes of operation, including a breakout detail view of the connection of the hydraulic actuator **410** to the support arm **132**/infeed frame **142**. FIG. **6** shows the cutter bar assembly in the rigid, raised position, FIG. **7** shows the cutter bar in over-traveled position, and FIG. **8** shows the cutter bar assembly in rigid raised position and over-traveled position in superimposition.

(31) Referring to FIG. **6**, the cutter bar assembly is shown in the raised rigid operating mode. In this mode, the hydraulic actuator **410** has extended fully as limited by actuator link **422** (not visible) and lifted the support arm **132**/infeed header **142** and the cutter bar **114** in the raised position relative to header frame **126** as the support arm **132**/infeed header **142** rotates about pivot **134**. As can be seen in the breakout, the extension of the hydraulic actuator **410** has forced pin **502** to the top of slot **500**, where the weight of support arm **132**/infeed header **142** and components attached to support arm **132**/infeed header **142**, e.g. the cutter bar **114**, infeed conveyor **120**, and related structures rests upon pin **502**.

(32) Referring to FIG. **7**, the cutter bar assembly is shown in the over-traveled operating mode. In this mode, the hydraulic actuator **410** has extended fully as limited by actuator link **422** (not visible) and lifted the support arm **132**/infeed header **142** and the cutter bar **114** in the raised position relative to header frame **126** as shown in FIG. **6**, but because of an obstacle **700** imparting an upward force to the support arm **132**/infeed header **142**, slot **502** is raised until pin **502** has stopped at the bottom of slot **502**. As can be seen in the breakout, the further rotation of support arm **132**/infeed header **142** about pivot **134** extension due to the obstacle **700** has forced pin **502** to the bottom of slot **500**, where slot **500** stops any further travel of support arm **132**/infeed header **142** beyond the rigid mode position.

(33) Referring to FIG. **8**, images of the cutter bar assembly in the rigid and over-traveled positions are superimposed, and the differences in positions of the components in the two modes can be seen. Between the rigid and over-traveled modes, pin **502** remains in place, but slot **500** rises from its position in the rigid mode until pin **502** reaches the lower limit of slot **500** as support arm **132**/infeed header **142** is rotated about pivot **134** by obstacle **700** and cutter bar **114** is raised.

(34) FIGS. **9**, **10**, and **11** show stop link **400** in various modes of operation. FIG. **9** shows the stop link **400** in the rigid, raised position, FIG. **10** shows the stop link in the fully-down position in flex mode, and FIG. **11** shows the stop link in the over-traveled position.

(35) Referring to FIG. **9**, stop link **400** is joined to header frame **126** at a first end **900** by a pivot **902** and at a second end **904** to the support arm **132**/infeed frame **142** by a pin slot joint formed by a slot **906** in the stop link **400** and a pin **908** fixed to the support arm **132**/infeed frame **142**. In an alternative arrangement (not shown), the pivot joint and the pin slot joint on stop link **400** are reversed, such that the stop link **400** is joined to the header frame **126** by a sliding joint and to the support arm **132**/infeed frame **142** by a pivot joint. Pivot **902** (or in the alternative, the sliding joint joining stop link **400** to header frame **126**) is located distal from and below the pivot **134** (not shown) joining the support arm **132**/infeed frame **142** to the header frame **126**. Slot **906** is sized such that when the support arm **132**/infeed frame **142** is in the raised position in the rigid mode, pin **908** has not reached the end of slot **906**. Thus the stop link permits the support arm **132**/infeed frame **142** to move upward, rotating about pivot **134** attached to header frame **126**, beyond the position in the rigid mode when support arm **132**/header frame **142** is subjected to a lifting force due to, for example, striking an obstacle while the cutter bar assembly is in rigid mode, until the pin **908** reaches the end of the slot **906**.

(36) Referring to FIG. **10**, stop link **400** is shown in the fully-down flex mode position. In this position, hydraulic actuator **410** is deactivated, allowing support arm **132**/infeed frame **142** to rotate downward about pivot **134** (not shown) until pin **908** reaches the end of slot **906** proximate to header frame **126**. In this position, support arm **132**/infeed header **142** may move freely within the limits of travel of pin **908** in slot **906**.
(37) Referring to FIG. **11**, stop link **400** is shown in the over-traveled position. In this mode, the support arm **132**/infeed header **142** and the cutter bar **114** (not shown) have been raised position relative to header frame **126** in the rigid mode, but an upward force to the support arm **132**/infeed header **142** has moved the support arm **132**/infeed frame **142** past the rigid raised position. The further rotation of support arm **132**/infeed header **142** about pivot **134** (not shown) has moved pin **908** to the end of slot **906** distal from header frame **126**, where slot **906** stops any further travel of support arm **132**/infeed header **142** beyond the rigid mode position. Stop link **400** can be provided with a strain gauge sensor or a load cell sensor (not shown) for detecting when the support arm **132**/infeed frame **142** reaches the over-traveled position and signalling a header position control system to adjust the header **112** position accordingly. Stop link **400** can further incorporate a spring (not shown) to allow the support arm **132**/infeed frame to move into the over-traveled position from the raised rigid position without damaging any components.
(38) In FIG. **12**, a position sensor **1200** is connected to the support arm **132**/infeed frame **142** by link **1202** at pivot **1204** for detecting when the support arm **132**/infeed frame **142** is rotated about pivot **134** from the raised position in the rigid mode by a lifting force at the distal end of the support arm **132**/infeed frame **142**. The position sensor **1200** can signal a header position control system when the support arm **132**/infeed frame reaches the over-traveled position to enable the header position control system to adjust the header **112** position accordingly.
(39) An example of a header position control system **1300** is shown in FIG. **13**. The control system **1300** may be implemented using any suitable arrangement of processors and logical circuits. FIG. **13** is a block diagram of exemplary hardware and computing equipment that may be used as a control system **1300**. The control system **1300** includes a central processing unit (CPU) **1302**, which is responsible for performing calculations and logic operations required to execute one or more computer programs or operations. The CPU **1302** is connected via a data transmission bus **1304**, to sensors **1306** (e.g., position sensors **1200** or sensors on stop link **400**), an input interface **1308**, an output interface **1310**, and a memory **1312**. The input and output interfaces **1308**, **1310** may comprise any suitable user-operable and perceivable system, such as a touchscreen controller/display, control knobs or joysticks, and the like. One or more analog to digital conversion circuits may be provided to convert analog data from the sensors **1306** to an appropriate digital signal for processing by the CPU **1302**, as known in the art. The CPU **1302** also may be operatively connected to one or more communication ports **1314**, such as serial communication ports, wireless communication ports (e.g., cellular or Bluetooth communication chipsets), or the like.
(40) The CPU **1302**, data transmission bus **1304** and memory **1312** may comprise any suitable computing device, such as an INTEL ATOM E3826 1.46 GHz Dual Core CPU or the like, being coupled to DDR3L 1066/1333 MHz SO-DIMM Socket SDRAM having a 4 GB memory capacity or other memory (e.g., compact disk, digital disk, solid state drive, flash memory, memory card, USB drive, optical disc storage, etc.). The selection of an appropriate processing system and memory is a matter of routine practice and need not be discussed in greater detail herein.
(41) Embodiments may be provided in various forms. In one instance, an embodiment may comprise an entire vehicle and header assembly, and the cutter bar assembly may be integrated into the header or into the vehicle. In another instance, an embodiment may comprise a header and an associated control system. Other configurations may be used in other embodiments.
(42) The present disclosure describes a number of inventive features and/or combinations of features that may be used alone or in combination with each other or in combination with other

technologies. It will be appreciated that embodiments may include any combination of support arms and associated linkages. For example, a header may have a combination of assemblies as shown in FIGS. **4** through **12**. One or more conventional support arms also may also be used in combination with one or more embodiments such as those described above. Other alternatives and variations will be apparent to persons of ordinary skill in the art in view of the present disclosure.
(43) The embodiments described herein are all exemplary, and are not intended to limit the scope of the claims. It will also be appreciated that the inventions described herein can be modified and adapted in various ways, and all such modifications and adaptations are intended to be included in the scope of this disclosure and the appended claims.

Claims

1. A cutter bar assembly for an agricultural vehicle header, comprising: a header frame; at least one support arm extending from the header frame in a forward direction, the support arm joined to the header frame at an end proximal to the header frame by a pivot; a cutter bar attached to a respective distal end of each at least one support arm; a stop link comprising first and second ends, wherein the first end joins one of the header frame or the support arm by a pivot and the second end joins the other of the header frame or the support arm by a sliding joint, and wherein the stop link joins the header frame at a point distal from the pivot joining the support arm to the header frame; and a hydraulic actuator joined at one end to the header frame by a pivot and at an opposite end to the support arm by a sliding joint; wherein the actuator can lift and hold the support arm in a raised position in a rigid mode and wherein the sliding joint joining the hydraulic actuator to the support arm and the stop link enable the support arm to move upward from the raised position in the rigid mode when the support arm is subjected to a lifting force at its distal end.
2. The cutter bar assembly of claim 1, wherein the at least one support arm comprises an infeed frame.
3. The cutter bar assembly of claim 1, wherein the sliding joint of the stop link comprises a pin slot joint comprising a slot in the link and a pin fixed respectively to the header frame or the support arm.
4. The cutter bar assembly of claim 1, wherein the stop link is joined by a pivot to the support arm and by a sliding joint to the header frame.
5. The cutter bar assembly of claim 1, wherein the stop link is joined to the header frame at a point below the pivot joining the support arm to the header frame.
6. The cutter bar assembly of claim 1, wherein the sliding joint joining the actuator to the support arm comprises a pin slot joint comprising a pin fixed to the actuator and a slot fixed to the support arm.
7. The cutter bar assembly of claim 1, further comprising an actuator link joined to the header frame and support arm in parallel with the hydraulic actuator to limit travel of the hydraulic actuator.
8. The cutter bar assembly of claim 7, wherein the actuator link is joined to the header frame and support arm by at least one sliding joint.
9. The cutter bar assembly of claim 8, wherein the sliding joint comprises a slot in the actuator link and a pin fixed respectively to either of the header frame and the support arm.
10. The cutter bar assembly of claim 1, wherein the stop link comprises a spring.
11. The cutter bar assembly of claim 1, wherein the stop link comprises a strain gauge sensor or a load cell sensor for detecting when the support arm is subjected to a lifting force at its distal end.
12. The cutter bar assembly of claim 1, wherein the support arm comprises a position sensor for detecting when the support arm is moved upward from the raised position in the rigid mode by a lifting force at its distal end.
13. An agricultural vehicle header, comprising the cutter bar assembly of claim 1.

14. The header of claim 13, comprising a header position control system.
15. The header of claim 14, wherein the cutter bar assembly further comprises a position sensor for detecting when the support arm is moved upward from the raised position in the rigid mode by a lifting force at its distal end and signalling the header position control system to raise the header.
16. The header of claim 13, wherein the stop link comprises a strain gauge sensor or a load cell sensor for detecting when the support arm is subjected to a lifting force at its distal end and signalling the header position control system to raise the header.
17. An agricultural combine harvester, comprising the header of claim 13.
18. A cutter bar assembly for an agricultural vehicle header, comprising: a header frame; an infeed frame extending from the header frame in a forward direction, the infeed frame joined to the header frame at an end proximal to the header frame by a pivot; a cutter bar attached to a respective distal end of the infeed frame; a stop link joined by a pivot to the infeed frame and by a pin slot joint to the header frame, wherein the pin slot joint comprises a slot in the stop link and a pin fixed to the header frame at a point below the pivot joining the infeed frame to the header frame; and a hydraulic actuator joined at one end to the header frame by a pivot and at an opposite end to the support arm by a sliding joint comprising a pin slot joint comprising a pin fixed to the actuator and a slot fixed to the infeed frame, wherein: the actuator can lift and hold the infeed frame in a raised position in a rigid mode; the sliding joint joining the hydraulic actuator to the infeed frame and the stop link enable the infeed frame to move upward from the raised position in the rigid mode when the infeed frame is subjected to a lifting force at its distal end; the cutter bar assembly further comprises an actuator link joined by sliding joints to the header frame and infeed frame in parallel with the hydraulic actuator to limit travel of the hydraulic actuator, wherein the sliding joints joining the actuator link to the header frame and infeed frame comprise a slot in the actuator link and a pin fixed respectively to each of the header frame and the infeed frame; and the cutter bar assembly further comprises a position sensor for detecting when the infeed frame is moved upward from the raised position in the rigid mode by a lifting force at its distal end.
19. An agricultural vehicle header, comprising the cutter bar assembly of claim 18 and a header position control system, wherein the infeed frame position sensor signals the header position control system to raise the header when the position sensor detects the infeed frame is moved upward from the raised position in the rigid mode by a lifting force at its distal end.
20. An agricultural combine harvester, comprising the header of claim 19.